

Leader's Report on XXV Indian Scientific Expedition to Antarctica—An Overview

L. Prem Kishore

Leader XXV ISEA

"Antarctica is suspended between two ages – those of competitive nationalism and cooperative internationalism". — Alan K. Herikson.

Antarctica continent, comprising almost 10% of the land surface of the globe, is linked to the vast extent of the Southern Ocean surrounding it. Remote from all centers of population it has no indigenous people living there. Antarctica the last heritage of mankind and the continent of extremes, has come to be known as the "Continent of Science". The white desert Antarctica is a pristine and natural laboratory, enabling the scientists to study, detect and monitor global phenomena, such as those related to atmospheric, meteorological parameter, oceanographic and glaciological, geological and geophysical research which may provide clue to the evolutionary history of the continents and the Paleo-climate of the earth.

Patterns of Antarctica Science and Adventure had been set in the early eighties in the context of Indian Antarctic Programme. Our continuous pursuance of science has been the driving force for the successful launching of the annual scientific expeditions to Antarctica every year since 1981 and the maintenance of year round permanent station Maitri is the testimony to the India's commitment to Antarctic Research. Indian Antarctic Programme has marched towards the Silver Jubilee expedition and it is a matter of happy coincidence that the silver jubilee celebrations of Indian Antarctic programme coincides with the International Polar Year Campaign.

Indian Antarctic Programme is multi-disciplinary and multi-institutional involving major themes focused on Meteorology and Atmospheric Sciences, Earth Sciences and Glaciology, Biology and Environmental Sciences, Human Physiology and Medicines and Engineering and Communications and Polar Logistics.

Humans do not belong to Antarctica. They go there like Astronauts entering deep space, taking with them everything needed for survival, i.e. accommodation, food and fuel. Every visitor is a transient, staying for limited periods, a few weeks or months, or wintering for a whole year. No

one lives indefinitely in Antarctica. Each expedition not only works in and copes with a mercilessly hostile environment but has to endure raw realities of human interaction in isolation, with attendant passions, personal idiosyncrasies, occasional medical emergencies and some grinding routine. The evolving importance of Antarctic Science is the changing emphasis from exploiting resources to conservation and new awareness of the importance of Antarctica in assessing global climate change.

Launching

The members of the 25th Silver Jubilee Indian Scientific Expedition to Antarctica assembled at National Centre for Antarctica & Ocean Research (NCAOR), Goa on 16th December 2005. A hearty and warm welcome was given by Dr S. Rajan, Dr M. Sudhakar, and Dr N. Khare. The team comprised of few experienced members of the past expeditions, budding enthusiastic youngsters and a contingent of Officers and other ranks from the Indian Army with the spirit of the expedition, zeal and enthusiasm. The same day the Insurance and other formalities were completed and the next four days up to 20th December 2005 were very hectic for the members with many activities, orientation lectures and the Launching. A chronology of the happenings, activities and events till the sailing is briefly given below.

A visit to M/s Sea Scan Marine Services Pvt. Ltd. Panaji was organized for giving essential training in Fire Fighting, Fire Prevention and Personal Survival Techniques. Both theories as well as practical demonstrations were given. This is very important for the members who venture out for an expedition to Antarctica where fire hazards are the foremost concern and principal danger at all Antarctic stations. There are detailed established procedures for fire prevention and fire fighting in Antarctica which every participating member should be well versed with.

Orientation lectures by eminent personalities on various topics were organized. Lectures on Utilities and up-keep of Maitri Station by Dr. S. Mukerji, Medical care by Dr. S.C. Tiwari, Environmental protocols by Shri Rasik Ravindra and Psychological aspects by Dr. Manju Mehta were action-packed and informative. These lectures provided the members the required knowledge, the inspiration and confidence to take up the challenges and tasks ahead of them.

Necessary instructions were given by Dr. N. Khare and Mr. P. Shetkar on various aspects of the submission of data acquired during the expedition at Antarctica to the National Antarctica Data Centre.

19th Dec. 2005 was the most memorable day for everybody present in the premises of NCAOR as the momentous Silver Jubilee Expedition of 25th ISEA was launched by Rear Admiral A.K Handa, Chairman & MD Goa Shipyard Ltd, as the Chief Guest; Shri Das, Adviser, DOD, Shri Rasik Ravindra, Dr. M. Sudhakar, Dr. S. Rajan, Dr. N. Khare and other dignitaries, Guests, Staff of NCAOR.

On 20th Dec. 2005 the members of the 25th ISEA were given a warm send off with blessings and good wishes by the Officials of NCAOR from Goa on way to Mumbai – Johannesburg – Cape Town.

On 21st Dec. 2005 the team arrived at Johannesburg and after immigration and custom clearance shifted from International terminal to the domestic terminal for onward journey to Cape Town.

23rd Dec. 2005 all the members assembled in the premises of the Indian High Commission, Cape Town where Dr. P.S. Goel, Secretary, Department of Ocean Development (now Ministry of Earth Sciences), Govt. of India and Mrs Banashri Bose In-charge of the Indian High Commission, Cape Town, addressed the expedition and gave assurance of all possible help to the expedition at any point of time in spite of the distance and problems of communication. This provided a big morale booster for the members to take up the challenges and assignments ahead of them.

24th Dec. 2005 – 28th Dec. 2005 the Vessel “M.V. Paardeberg was taken over for the 25th ISEA after completing the formalities the members shifted on board. Shri T.V.P. Bhaksara Rao made all the necessary arrangements, supervised the operations of loading of Piston Bully, Mantis Crane, Helicopters and the containers which had come from Goa, along with other logistics requirements onto the Vessel. A thorough checking of the materials was carried out which were loaded at Cape Town especially the spares, clothing and other essentials. All the members were issued clothing kit to be used during the expedition. Next couple of days up to 28th Dec. 2005 were spent in preparation of the vessel for the long journey to be undertaken in the most hostile conditions of Southern Ocean.

The Voyage

Every Expedition is equally challenging for the team and every Expedition has its own charm and challenges and a commitment. With this set of mind every member, of the 25th Silver Jubilee Indian Scientific Expedition to Antarctica, sailed off from Cape Town at 08:30 hrs. On 29th December, 2005 on board the Vessel “M.V. Paardeberg”. The team comprised of 52 members, 27 members for the summer and 25 for the winter. The team comprised of 34 Scientists from 18 Organizations/ Academic Institutes, 14 members Logistic Team from Indian Army and 2 cooks from Indo Tibetan Border Police. Two Medical Officers also joined the expedition as winter members.

The journey was smooth for a couple of days but on 2nd Jan. 2006 some unforeseen problems encountered with frequent breaking of the pipes leading to the boiler and a good amount of fresh water supply at the vessel was drained out. This created severe shortage of fresh water on board. With the sudden breakdown there was some inconvenience and apprehensions in the minds of everybody including the Captain, Ice Pilot, Engineers, the crew and members for the onward journey to take place but with the timely intervention and experience of all concerned the problems was sorted out, the repairs works taken up and the journey proceeded further. Things were brought to normalcy onboard the ship in spite of some shortage of the fresh water.

During the remaining part of the journey necessary plan of work during the summer time were discussed and finalized. Also details were worked out for the transportation of members and the equipment first to the shelf and subsequently to Maitri and other sites for starting the activities in the respective fields. For this detailed discussions were held with the members and the Helicopter crew. An unambiguous picture of the operations to be undertaken especially shifting of man and machinery to their respective destination safely was given to the crew. In fact the whole expedition team and other support staff got acquainted with the entire programme of work during the expedition. This was very important as all team members were from different organizations and diverse backgrounds.

The journey also gave an opportunity for the members to witness and experience the roaring Forties, furious Fifties and screaming Sixties, apart

from witnessing the beauty of the Southern Ocean with its splendid Flora and Fauna. Albatross, Snow Petrels and other birds having a smooth flight in the open sky, Dolphins dancing Whales with their furious dives into the ocean, waters Seals basking and the most exciting eye catching was the sighting of the Ice Bergs with different shapes and sizes a feast to watch for the human eye in the blue waters of the wilderness.

This is the time when one comes to know the real beauty of nature undisturbed, untouched, and quiet and in peace in its own way. With the vessel “M.V. Paardeberg” entry into the pack ice, the natives of Antarctica “Penguins” gave a warm welcome to the thrilling and excitement of everybody on board. With the latest technology “Digital Cameras” in the hands of members clicking was the order of the day throughout.

The Ship crossed the demarcation point of the Southern Ocean the Antarctic Circle on 8th Jan. 2006 at 08:40 hrs. and celebrations of the same was there on the Helicopter Deck of the Ship with stamping of seal on the face for the first timers with fun, fanfare, barbeque, music and dance.

In spite of some odds on board the vessel especially the technical snags which were developed during the journey and duly repaired, the vessel reached the fast ice i.e., the solid and stable ice on 9th January, 2006 from where Helicopters sorties could be operated.

10th Jan. 2006 was the most memorable day as a courtesy sortie was made to the abode of Maitri - a home away from home, where a traditional welcome was given to the new team by the inmates of the Maitri station from 24th ISEA, living confined in the solitary environment for over an year, who were eagerly waiting for our arrival. They were impatiently waiting for the fresh supply of vegetables and fruits and other essentials and gifts from near and dear ones from home. From the very next day onwards the activities started from the ship shifting of the members along with equipment and essentials to Maitri. First, the summer team members were shifted. By 14th Jan 2006 the summer camp was fully established. Since then all the summer activities were started with many of the members on fieldwork, data collection, observations as per their programme of work, tasks planned before hand. Shri Javed Beg was nominated the Summer Camp In-charge.

SUMMER ACTIVITIES

Scientific

The summer scientific activities started by 12th January 2006 with many of the members forming different field parties as per their nature of work like Survey, sample collections and data observations in and around the Schimarcher Oasis region. Following are the details of the various Organizations/Institutes and their activity.

National Centre for Antarctic & Ocean Research (NCAOR): The demarcation of lakes shoreline of the Priyadarashini Lake was carried out for hydrographic survey. Four sediment cores from different lakes were collected for paleo-climatic studies and also two sediment cores from Priyadarshini Lake. Bathymetric survey of Priyadarshini Lake using Echo sounder was carried out by Dr S.K. Chaturvedi.

Geological Survey of India (GSI): Data acquisition was carried out by using GPR instrument. Initial three Traverses, south of Schirmacher were devoted to testing of attachments/equipments and for calibration of the antennas. Later on, two traverses using 200 MHz and 100 MHz Bi-static antennas were taken between Nunataks Tallaksenvarden – Vitahia and Vitahia – Schirmacher. Another 10 km East – West traverse over the polar ice cap was taken up to the western most edge of the Schirmacher. Two 400 mts profiles; North – South and East – West using 200 MHz and 100 MHz Bi-Static antenna were taken up around Ice Core Drilling site at 71 deg 20' 38" S and 11 deg 35' 38" E for selecting a suitable site and to study the snow stratigraphy to ascertain the continuity sequence of the Ice layers. The equipment was also used for crevasse detection on way to establishing the Ice Core drilling camp about 70 km south of Maitri. The depth profiles with coordinate tags for almost three fourth of Zub (Priyadarshani) Lake using the Hydro Box and handheld Garmin GPS were carried out by Mr. Javed Beg.

Based on the data of detailed GPR study a drilling site was selected on Humboldt glacier on 71 deg 20' 38" S and 11 deg 35' 38" E. Ice core output of 65.04 meters was recovered from the main borehole. Another supporting borehole of 3.5 meters was made to assess the top layers. Additionally, a trench was made using a chain saw and bulk sampling of snow and firm ice was done from this pit by M/s Arun Chaturvedi, A. Rajkumar and P.C. George.

NCAOR-GSI:

Ice Core Drilling operation: Drilling operation continued from the month of November 2006 on the Ice Shelf of East Antarctica (Lat 70° 11' 57.4" S and Long 12° 27' 25.1" E). Drilling reached 74.98 Mts and drilling operation had to be terminated due to the extreme difficulties in recovery and the fragile nature of the cores collected. All the ice cores have been packed properly in LDPE bags and safely placed in EPP boxes for further transport to Goa. Drilling camp closed down and another reconnaissance survey was carried out in the ice shelf for the second site of drilling operation. The location has been finalized with Lat 70° 13.804' S and Long 11° 55.738' E.

Field work in Schirmacher Oasis, Veteheia Hills and Veten (Shivling) Nunatak: Field traverse and field work was carried out for 3 field days in Schirmacher Oasis. The objectives were (a) To locate the major epishelf lakes and to plan a strategy to carry out sampling for dating and reconstructing the history of possible push moraine deposits/sea level changes. (b) To identify the geological structures associated with the country rock and intrusive.

Traverse and field work on the Veteheia hills and Veten (Shivling) Nunatak was carried out for 7 field days. The aim was to identify the structures associated with the country rock and the intrusives as well as for collecting samples for further study. Thereafter, data plotting and literature consultations were made at the camp.

Physical verification of DOD/NCAOR assets: As entrusted by NCAOR, an exhaustive physical verification process was attempted on the assets as per the list provided by DOD (presently Ministry of Earth Sciences). Many of the items were identified after painstaking field surveys. The job is an ongoing process. In spite of spending lots of time and effort, comprehensive and final assets verification seems to be extremely difficult due to manifold reasons that would be described in a detailed manner in the final report on the above.

Sampling along the periphery of north Humboldt pluton and the Zweissel pluton was done. Totally ten-drop points were selected after carefully studying the plutons and sampling was done for detailed geochemical and petrochemical studies. 55 rock samples each weighing

more than 15 kgs were collected, as the rock was very coarse-grained granite, gabbros, charnockiteetc, (Mervin D'Souza).

Space Application Centre (SAC): In-house developed instruments were used for ground truth measurements. Narrow band Ground, Broad Band, Near Infrared and Ration Radiometer modules were used. A test site for vicarious calibration of IRS P6 AwiFS sensor was selected at Sankalp and Airport as the area remains covered with uniform ice throughout the year, except during December and January. Ground truth data collection and measurements were conducted in terms of the following parameters.

- i. Reflectance measurement of snow surfaces around Maitri Station.
- ii. Reflectance measurement of ice and fresh snow surfaces around Sankalp area /Vatiayah Hill.
- iii. Terrestrial solar radions: Irradiance and Polarizations.
- iv. It has been observed that the data collected was consistent.

Indian Institute of Tropical Meteorology (IITM): To study the direct irradiative forcing due to Aerosols and precursor gases over Antarctic region.

Investigated the optical micro-physical and irradiative properties of aerosols including their irradiative forcing and heterogeneous chemical effects, by using spectral solar radiometers and wideband short wave (SW) payronometer. It has been observed that the parameters showed considerable variations in daily mean values of AOD and Column water vapour.

Visits of Foreign Delegates and Members from other Countries

With establishment of Air Link from Cape Town to Novo Air base (Russian) operated by ALCI (Antarctic Logistics Centre International) and onward transportation of members from different countries to their respective summer camps for field, Maitri station has been attracting number of visitors from different nation to have first hand information about the ongoing programmes at the Indian base station.

National Geophysical Research Institute (NGRI): Permanent Seismic and GPS stations are under operation at Maitri. The recording continued during the expedition. Downloading and analysis of the data

was carried out daily and results were communicated to Hyderabad. The underground Seismometer vault constructed in the year 1997 was opened and maintenance of the pier was carried out for the installation and up gradation of Broad Band system. Test recording was carried out by M/s L. Prem Kishore and S. V. R. Ramachandra Rao.

Magneto telluric (MTS) investigations were carried out by M/s S. Prabhakar, E. Rao and Sarana Basavapa in Schirmacher Oasis and surrounding areas. Data have been collected at 6 (six) MT sites to the south of Maitri Station with station interval of 3 – 4 km in the frequency range of 1000 Hz to 0.0001 Hz, for a period of 4 – 5 days per site.

Survey of India (SOI): 30 days of GPS observation was made by M/s Anurag Sharma and Anil K Mehta. The data was collected and stored for further processing. Identification and Re-observation of 28 GPS stations and the data was archived for final processing at Head Quarters. Detailed mapping of approximately 2 sq km area was completed.

Birbal Sahani Institute of Palaeobotany (BSIP): Thirty lakes were surveyed and water samples were collected for analysis of different parameters like temperature, pH, TDS, EC and salinity. Ten number of sediment cores were collected and sub sampled for Paleo climatic studies. About 200 sediment samples have been collected. The geomorphological data has been collected in the surveyed area and aerial photography of lakes in the Schirmacher Oasis was done. (Dr. Binita Phartiyal)

Centre for Development of Advanced Computing (CDAC): 150 GPS points were established across different parts of Schimacher Oasis covering areas between Novo and DG Snout, Long Lake behind Trishul point. These points will be used as ground control points for geo-referencing of Satellite Image to prepare a base map.

National Hydrographic Office (NHO): The team consisting of Lt.Cdr J. Gurumani, PO Yashpal Kalra and LSSR-1 Vinod Kumar carried weather observation from Cape Town to Indian Bay during the voyage. The team recorded bathymetric data manually when the ship was steaming within the survey area. During the stay at India Bay weather observations were recorded at synoptic hours.

Snow and Avalanche Study Establishment (SASE): A tower was installed on the shelf ice. One RRTS (Remote relay telemetry system) weather station was established on the tower for recording the weather parameters on to a data logger by Mr. Manoj Kumar.

Indian Institute of Geomagnetism (IIG): The geomagnetic data was collected by Mr. Atul Kulkarni continuously on Rio meter and magnetometers in analog and digital mode and the total magnetic field intensity (F) using Proton Precession Magnetometer and three components X, Y, Z using digital Fluxgate Magnetometer. A passive antenna for measuring atmospheric potential was installed. One GPS unit is installed for continuous monitoring at Maitri, the data will be used for atmospheric studies of measurement of TEC, scintillation and water vapour content.

India Meteorological Department (IMD): Synoptic observations were made daily of atmospheric pressure, surface wind, surface air temperature and cloud cover. These observations were made after three-six hour observations (00, 06, 12 & 18 hrs.) and were transmitted on real time basis to IMD New Delhi. Radiation studies for Diffuse and direct solar radiation was monitored continuously. Balloon borne radiometer sonde was sent every fortnightly after sunset to study short and long wave radiations in the atmosphere. Ozone sonde instrument was released every week. Atmospheric turbidity was measured with the help of SUN Photometer every 2 hrs. Brewer Spectrometer was operated on good sunshine days.

Barkatullah University: During onward voyage (onboard ship) with the help of a GPS system ionosphere monitoring observations were carried out. At Maitri a GPS station was established at location $70^{\circ} 46' 1.7''$ S and $11^{\circ} 44' 7.4''$. For monitoring Scintillation and Total Electron content, data was collected up to 1363 GPS week along with these computed records of inertial navigation system, ionospheric correction, ionospheric delay (cm), ionospheric rate (0.05 cm/min) were collected to study the ionospheric conditions under the influence of geomagnetism. Along with this raw data, time, position and fix related data of the GPS receiver and data of tracked satellites was collected. Vivek Mishra collected water samples from different lakes of Schirmacher Oasis.

National Physical Laboratory (NPL): For the study to probe vertical structure of the atmospheric boundary layer over Maitri, Antarctica a mini acoustic sounder was inducted by Dr. H.N. Dutta. The equipment was installed and tested for its data quality near NPL hut. The data probing up to a height of 640 meters in altitude with a sweep rate of 8 seconds was collected automatically on a PC.

Botanical Survey of India (BSI): Samples of Algae were collected from different lakes in Schirmacher oasis region by Dr. M Palanisamy, (total 102 samples in the first phase and 32 algal samples in the second phase in different lakes of Oasis). The collected samples were preserved with 4% formalin for further study at BSI Lab at Port Blair.

Nicholas Piramal Research Centre (NPRC): Mr. Shailendra K. Sonawane collected 60 Soil/organic samples from different sites from Schirmacher Oasis region. He also collected water samples from 10 different lakes plated on nutrient agar, actinomycetes agar and luria broth for bacterial growth.

Defense Electronic Application Laboratory (DEAL): Mr. Jagmal Singh organized setting up of communication system on the Vessel "M.V. Paardeberg" which included installation of HF and VHF equipments and Antennas for communication between ship and Maitri and field parties. Fabrication and installation of Log Periodic Antenna for HF communication near the main structure of Maitri station. VHF repeater and antennas at Vetiah top were also installed. An e-mail service on INMARSAT mini-M (Nera) terminal was also configured. Ground truth confirmation facility in the Schirmacher Oasis for SAR image interpretation was also set up.

Defense Institute of Psychological Research (DIPR): Preliminary data from 25th Expedition members on Psychological parameters have been collected by Dr. M.R. Panwar. A few psychological tests were also conducted on the members of scientific team of 24th Expedition members to study post expedition influence.

National Botanical Research Institute (NBRI): Identified the habitats of Lichen growth with respect to glacier and distribution of 2 species was monitored. Observation on the relations of lichen growth and rock type was done. Pilot experiments on field observations of lichen ecology were carried out by Ms Preeti Oswal.

MES Abasaheb Garware College (MES): Mr. Ashok Survade collected 22 samples of Penguin fecal matter on fast ice. Skua fecal matter sample in and around Schirmacher Oasis were also collected. Ten Skua nests were sighted and behavioral observation of Skua was monitored throughout the period of stay.

Logistics

The Indian Army Logistic Team consisting of 2 Officers, 5 JCOs and 6 other ranks reached the Indian Antarctic Station Maitri on 15th Jan. 06. The team was involved in a number of tasks as part of the logistic support to the expedition. The detailed report of the tasks carried out is enumerated in the subsequent paragraphs.

Cargo Handling

The most important task on arrival at Antarctica was to shift the scientific and logistic stores from the ship to the station. To accomplish this, the stores loaded in the ship MV Paardeberg were taken stock of and segregated during the sailing period according to their priority in urgency requirement at Maitri. Part loads were prepared according to the weight and volume of the chartered helicopters capacity. The essential winter clothing required by the members of the team was issued along with other clothing on 01, 02 Jan. 2006 at Cape Town itself by NCOAR. These segregated stores were then transported by helicopters to Maitri. Remaining stores were transported later through land convoys.

Handing and Taking over stores and materials

As soon as the team reached Maitri all personnel were directed to liaise with their counterparts of the 24th ISAE. Handing and taking over of the station and training of the 25th winter team on equipment and their operation was initiated.

Up-keep of Maitri summer camp

Accommodation: Bath modules in the summer camp were repaired. It is customary to repair the Maitri station structures quite often as they are semi-permanent beside heavy winds and extreme cold affects the structures.

Water Supply: Additional insulation and heating was provided for uninterrupted hot water supply.

Heating: Electric load was adjusted in various distribution circuits to accommodate more heaters and thus a better standard of living could be achieved.

Civil Construction Jobs: As part of logistic support for carrying out scientific experiments, the following tasks assigned have been completed.

As desired by NGRI, the Construction of a Concrete pillar with a stable platform and to house the Short Period, Broad Band Seismometers was taken up.

Two 8 mtr high antennas of SASE were erected one each at the Indian bay on the shelf and on cap ice near Banjara point.

The high frequency antenna of DEAL was erected at the station.

The huge heap of rocks and boulders has been cleared to prevent accumulation of snow around the generator complex especially during winter period.

The plinth around the A block has been further strengthened by dozing sand and mud under the super structure sealing it off with cement concrete.

A few spots snow ingress were identified in the A block and were sealed to prevent further damage.

Boiler Room and Central Heating System is one of the life supporting systems of Maitri Station. The new monobloc pumps received from NCAOR were put into operation by replacing the defective older pumps for circulation of hot water through the station. However, these pumps were getting overheated as they are basically designed for cold water and not hot water. The pumps were then relocated before the damage could occur and were made operational. It may be essential to install pumps suitable for hot water.

Management of waste: Due to strict environmental regulations the management of waste is a very critical operation in Antarctica. Therefore due importance is being laid to this effect and the following works have been carried out.

The soak-pit was cleaned and its depth was increased with the help of dozers to accommodate more volume of waste water and thus reducing the

risk of water over flowing to the Priyadarshini Lake, which is the source of drinking water.

Both, the urinal as well as the kitchen Klargesters were drained using improvised pump-cum-strainer and cleaned.

Material that is detrimental to the Environment, as per the CEP guidelines, was stored into the back loading containers, which were back loaded from the station during the return convoy.

As part of the cleanliness drive, the station was cleaned up with the participation of the entire team and the combustible waste was processed in the incinerator. All the containers with old stocks have been readjusted to cater for the new stocks due to arrive in the convoy.

Power Generation

Power Generation is the heart of Maitri and is critical for all the life support systems at Maitri. The following tasks were carried out to ensure uninterrupted power generation:

A 62 KVA generator was repaired by replacing the voltage regulator module and made functional.

The cooling system of a 125 KVA generator was kept under observation and rectification through a trial and error to avoid over heating of the engine especially during summer period.

Fire fighting

All unserviceable fire fighting equipment were replaced with functional ones.

A briefing on fire fighting procedures was organized by the fire officer and the drills were practiced.

The automatic fire alarm system of the station was tested for functionality.

Maintenance and operation of Snow-Vehicles Cranes and Bull dozers

Convoys are the lifeline of any expedition in Antarctica as all supplies to the station depend on it. To keep all the vehicles in running condition is the arduous task assigned to the Logistic team and the same were carried out meticulously.

A Piston Bulley Vehicle, which was off road for want of an engine assembly, was made functional by replacing it with a new engine assembly.

The Libeherr Crane, which were off road was repaired and made operational.

All container-towing trailers were checked for their serviceability and kept ready for the first convoy by replacing axles and the tracks on the trailers as most of the tyres/wheels of the trailers have out lived.

Entire workshop area was cleaned; unserviceable spares and scrap have been put into the Back loading container.

A Ramp for inspection of vehicles outside the workshop shed was planned for construction by using the scrap material available in the station.

The track and the dozing blade/bucket have been mounted on the newly received Piston Bulley-300 vehicle. The new Piston Bulley-300 and the 18 ton Mantis Crane 3612 were tested for their operability.

Scientists from the GSI, DEAL, SASE and NGRI (MTS group) were ferried, across inhospitable terrain, in the Piston Bulley Vehicle to accomplish their Scientific Tasks. One piston Bulley vehicle was modified to suit the fitment of GPR antenna system of GSI to carry out their studies.

Ice Core Drilling Camp

On 29th Jan. 2006 a convoy of 5 Piston Bulley vehicles with all the relevant stores for the drilling camp led by Lt. Cols. R. S. Sial, R. Janardhana Rao along with Shri Arun Chaturvedi and his Ice Core drilling team were moved to the drilling site through a severe blizzard under near zero visibility conditions. The camp was established by 1st Feb. 2006 and the convoy returned on the same day, through yet another day of bad weather with heavy winds at 40 knots and severe drifting of snow. The camp was to be wound up on 15th Feb. 2006 with severe weather conditions looming large over entire area of operation a convoy of lesser size with minimum number of vehicles and lesser support involving a high degree of risk was planned. The convoy led by Lt Col R Janardhana Rao left for the drilling camp and returned back after having wound up the drilling camp, within a span of 18 hours covering a distance of over 150 Kms through a very severe blizzard with zero visibility conditions. The drilling team was brought back to the station safely.

Closure of Summer Camp

The summer period activities of 25th ISEA were closed on 15th February, 2006 as per schedule. The induction of the summer members onto the ship started on 7th February and by 20th February depending upon the operation of Helicopters all the summer members were on board the ship except for Mr Javed Beg, who accompanied with the convoy for the GPR experiment.

The first phase of the Scientific Objectives envisaged for the 25th ISEA was successfully accomplished with majority of the members able to collect the required samples, make observations and collect valuable information in the form data for further analysis, interpretation and to delineate the valuable information to further enhance the knowledge of Polar Science. These things could be achieved in spite of many uncertainties like the problems encountered on board the ship, constraints of the Helicopter support in the mountain regions and the weather which plays an important role in the harsh and hostile conditions of Antarctica. All the members have made use of the best working days and good weather conditions in the stipulated time period for fulfilling their objectives and tasks.

The formalities of handing and taking over from 22nd wintering over team to 23rd wintering over team were completed on 21st February 2006, a joint convey of 24th and 25th ISEA was done to the shelf and with this all the members of 24th ISEA were on board the ship. The operations on the ship of decanting, cargo offloading have been executed by 26th Feb. 06 and after the preparedness of the ship the return voyage of the 24th ISEA started on 4th March 2006.

Wintering Over

The wintering team of 25th ISEA after the completion of the formalities of handing and taking over of the station, was enthusiastic in carrying out the programmes, tasks and other assignments for headway in science and crucial polar logistics. Winter programmes related to Geophysical, Atmospheric Meteorological, Glaciology, Human Physiology Communications, and Logistics were undertaken with the involvement of National Geophysical Research Institute, Hyderabad, National Centre for Antarctic and Ocean Research, Goa, Indian Institute of Geomagnetism, Mumbai, India Meteorological Department - Govt. of India, Snow and

Avalanche Study Establishment, Defense Institute of Physiological Research, and Defence Electronic Application Laboratory, Indian Army and Indo-Tibetan Border Police, etc. Every effort has been made to acquire valuable information, parameters and data to further pursuance of the Antarctic Science and has given out the best in betterment of the environment around Maitri station.

To make use of the summer daylight many activities of indoors and outdoors were taken up. After the successful return of the first convoy from the shelf on 1st March 2006, the successive convoys started bringing the various supplies of rations, medicines, fuel, spares and other essentials for the maintenance of the station and running of various systems for our stay. The team got busy in segregating the various items as and when the convoys returned back and arranging these stores in various containerized stores especially provisions, toiletries, spares and other essentials behind the station which were marked exclusively and stacked according to the requirement of the need as and when required, as well as the materials immediately required on the loft of the Kitchen. Some of the items were also sorted and stored in the container complex.

Simultaneously various ongoing scientific programmes from NGRI, IIG, SASE, IMD, DIPR and other logistics tasks were being carried out especially the outdoor activities as and when the weather was clear.

In order to facilitate the smooth operation and functioning of various systems and for carrying out various programmes, tasks and assignments envisaged for the 25th ISEA and in fulfilling the scientific objectives and logistics activities members were made in-charge of various activities.

Lt. Col. R. Janardhan Rao was nominated as the Deputy Leader of the 25th ISEA for assisting in planning, execution, station maintenance, environmental cleanliness, waste management and other assignments and tasks to be carried out during the expedition.

Polar Nights

Once the winter set in, most of the activities were confined to the indoors. This period was utilized for the maintenance of the station and the generators. Demand lists of required items for the coming expedition were prepared. All the medicines which accompanied the expedition and available already at Maitri were sorted out and arranged in a systematic way for easy

access. All expired medicines were cleared off. X-Ray, The Pathology and the OT room were reorganized as per the convenience of the Doctors for any emergency and eventuality.

Wintering over in Antarctica during the polar nights is a rare experience where you have a feeling of loneliness, boredom and long period of restricted activities outside the base station. In order to overcome all these, various sporting activities and competitions were organized and the mega serials of Ramayana, Mahabharata and Krishna were screened. To have a feeling of homely atmosphere and keep up the moral of the members high birthdays, wedding anniversaries, national celebrations and some of the important Antarctic events were celebrated.

The peak winter period was witnessed by a severe blizzard that continued for almost a month. All the members were on alert for any eventuality. The most hostile and harsh conditions are witnessed during the blizzards by which there is a huge accumulation of snow everywhere and it becomes difficult to run the systems, especially the generators, the life support systems, maintenance of water supply and various experiments and other scientific activities. In order to monitor and clearance of the accumulated snow various teams were formed for regular clean up on rotation basis and monitored the station for any emergency.

All efforts were made to maintain physical fitness of the team during the expedition. Accidently Nb Sub MG Chavan suffered an injury in left hand during one of the convoys. The JCO underwent a surgery at the Russian base and was rested for more than 5 months to completely recover. As every individual was conscious of Physical fitness and obesity, a spacious gym was created inside the station by realignment of the available space. The team members were kept busy with all sporting activities like table tennis and gym routine. There was voluntary participation in this regard. During the polar nights, Yoga was conducted to overcome psychological stresses that normally develop in such closed environment. Indoor game tournaments were also organized. Two cricket matches were played. Total 53 celebrations on the occasions of birthdays, anniversaries, national and religious festivals were organized. To carryout various activities at the Maitri station 27 'Shramdands' were organized where all the members voluntarily participated.

Experiencing the Antarctic winter is a very rare experience and witnessing the Auroras is one among them. Members braved the cold and kept awake many a nights to arrest the Aurora's in their cameras. The first sunrise made all the members to have a festive mood at the station. These are some of the memorable moments and events in an expedition which cannot be easily forgotten.

The first visitor to the Maitri station "Skua" brought back the movements of other living beings in the vicinity of the station and with the increase of the daylight day by day the outdoor activities were taken up especially the preparation of the back loading containers in the form of Shramdan's (voluntary service) by the members. This being a huge task lot of efforts had to be put in during the fair weather days. Preparation of the summer camp were carried out for the coming teams to have a comfortable stay during the summer time to carry out their respective assignments and tasks.

Aerial Visit to Vostok Station

The Indian and Russian Antarctic Programmes for the past few years are playing an important role in providing Logistic support to ALCI for flight operations at Novo Air base, facilitating movement of expedition members as well as cargo, both Scientific and Logistics, into CDML region of this white desert.

During the 25th ISEA, a substantial amount of logistic support was provided by our logistic team to the Novo Air base especially transportation of 80 KL of Aviation Kerosene for the IL-76 from the Russian Barrier in the hour of crisis, unloading of huge cargo from the aircraft and maintenance of the Runway. The ALCI management were surprised at the quick reaction of our logistic team in moving to the Novo air base and completing the snow clearance in a span of 3 hours from the time they made an S.O.S. call, facilitating departure of the IL-76 as per schedule with Cargo for Vostok.

Keeping in view of the above assistance rendered to the Novo Air Strip there was an invitation for 4 members of the Indian team from the Manager of the Novo Air Strip to take part in the operations of Para dropping of fuel and some essentials onto the Vostok Station. The following members have taken part, two members each for the two flight schedules of 30th November 2006 and 1st December 2006. The journey time is around 6 hrs

to and fro from Novo Air Strip to Vostok and back.

1. L. Prem Kishore
2. Lt. Col. R. Janardhana Rao
3. Lt. Col. Suresh Kumar
4. Capt. Kingston

Conceptual Changes in the Indian Antarctica Programme During the XXV ISEA

Shri Rasik Ravindra, Director, NCAOR has brought conceptual changes in the ongoing schedule programmes of Indian Antarctic Expeditions starting 25th ISEA. In order to make use of the best period of the summer time of the Antarctic weather conditions, induction of the next summer and winter members have taken place during the month of November 2006 itself. A total of 23 members along with an Engineer from Kashbohrer in two batches comprising 14 and 9 members arrived by making use of the air link from Cape Town to Novo Airstrip close by to Maitri Station.

Visit of Dr. P.S. Goel, Secretary, MoES

As an icing on the cake, the profound success of the Silver Jubilee expedition was marked by the visit of Dr. P. S. Goel, Secretary, MoES, along with Shri Rasik Ravindra, Director, NCAOR and Shri Prakash Kumar Joint Secretary, MoES to Maitri from 13th to 15th Feb. 2007.

The Secretary visited each and every corner of the Maitri Station, Laboratories and ongoing experiments being undertaken and other infrastructural facilities available at Maitri Station complex. He interacted with members and discussed in detail about their participation in the ongoing Scientific programs and Logistics tasks and appreciated the efforts put up by the members in augmentation of scientific data, the collection of samples and other related ongoing scientific programs and logistic efforts.

Dr. P. S. Goel took note of the difficulties and hardships faced by the members of the team in the harsh and hostile weather conditions

especially during the blizzards for the smooth functioning of the scientific instruments for collection of data and specially on convoys. He also appreciated the enthusiasm shown by the members in meeting their targets and completion of the assigned tasks and other related scientific programs, engineering efforts and innovations put in to maintain the generators, the boiler room and the modifications carried out on the generators to rectify the overheating problem and the manner in which the summer camp has been established.

Dr. P. S. Goel expressed satisfaction at the way the ongoing programs were initiated and completed as per schedule in spite of the difficulties faced by the members. Shri Prakash Kumar had lengthy discussion with all the members of the logistic team for obtaining the solutions for the existing logistic problems.

As a brand Ambassador of India Dr. P.S. Geol along with Shri Rasik Ravindra, Shri Prakash Kumar visited the Russian station and interacted with the Leader and other members of the Russian team. It was a surprise to all of us that the Secretary has made it a point to walk along the rough and hilly terrain consisting of boulders etc., towards the Russian station.

The story of the 25th ISEA combines courage, endurance, humor, beauty and drama exhibited by an extraordinary community of 25 members who have stayed here during the whole of winter in 2006, achieving prodigious accomplishments in the face of the extreme hazards and difficulties of the Antarctic environment. It was also a significant expedition as second time an Indian lady Dr. Deviyani Borole, Doctor, wintered over and braved all the extremities of the Icy Continent. The station was finally handed over to the new team headed by Shri Jayapaul of Geological Survey of India, on 17th Feb. 2007. The team returned to India on 21st Feb. 2007, ending yet another great Indian expedition.



Launching Ceremony at NCAOR, Goa Rear Admiral A.K. Handa (in the middle), Chief Guest, from left Shri Das, Shri L. Prem Kishore Shri Rasik Ravindra, Dr. M. Sudhakar



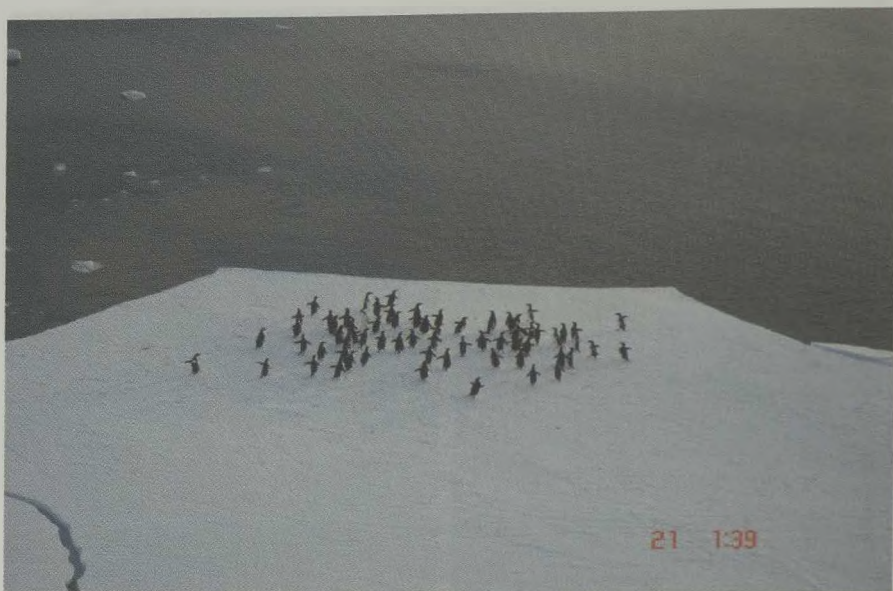
Team Members along with Dr. P.S. Goel and Mrs Banashri Bose In-charge, Indian High Commission, Cape Town



Loading of Piston Bully and Mantis Crane onto to the Ship at Cape Town Harbour



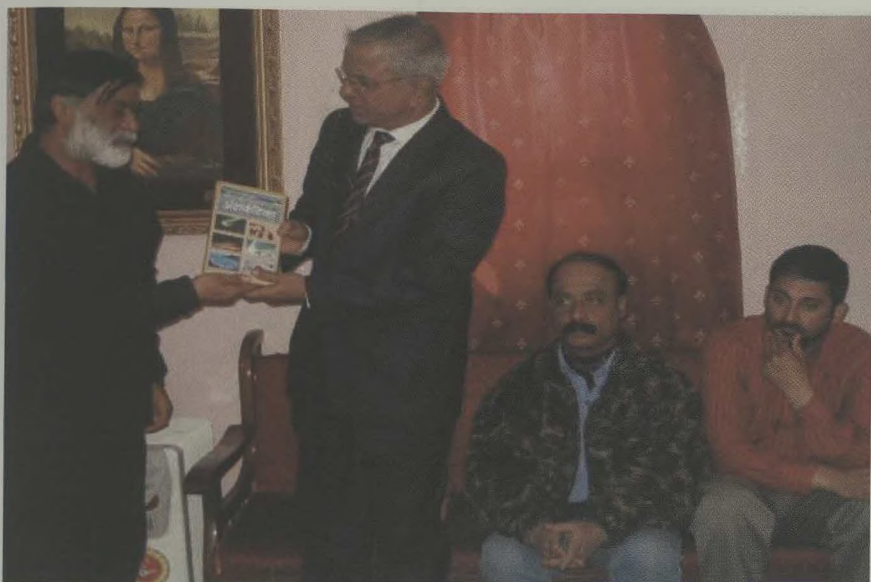
Camel shape Ice-Berg on way to Antarctica



Welcome at Antarctica by Penguins



Skua with its own image in clear waters



**Release of Book Titled “Antarctica” in Hindi written by
Shri Arun Chaturvedi (in left)**



**Handing over of the Maitri station by the outgoing Leader of 24th ISEA
Shri Rajesh Ashtana (in Jersey) to Shri L. Prem Kishore Leader of 25th ISEA,
Col. R. Janardhan Rao (extreme left), Col. Sial (extreme right)**



First Sun rise after the Polar Nights from Marutis



Arrival of Dr. P. S. Goel (in Red Jersey) Shri Rashik Ravindra (Brown Jersey) and Shri Prakash Kumar at Russian Airstrip



Dr. P. S. Goel along with Snow Cat Vehicles



Dignitaries inside the Leader Room



Inauguration of "Trisna" new Pump House by Dr. P. S. Goel



Visit to the Russian Station Novolazarevskaya. Dr. P.S. Goel along with Mr. Nikoloy (extreme left) Leader of Russia Station, Shri Rasik Ravindra (extreme right)



Scenic beauty of Tran's Antarctic mountain ranges, view from ALCI flight on way to Vostok



Handing over of Maitri Station to Shri Jaya Paul, Leader of 26th ISEA by Shri L. Prem Kishore, Leader of 25th ISEA, Col. Suresh (extreme left) Col. R. Janardhan Rao (extreme right)